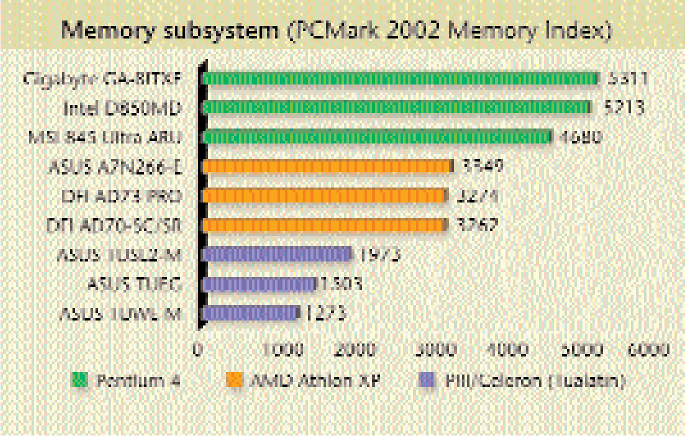

dle these variations and at the same time maintain accurate and tight timing cycles to ensure error-free and high-speed data throughput.

Pentium 4: It's really hard to compete with the RDRAM boards due to the raw memory throughput that RDRAM offers. The scores for both the Intel D850MD and the Gigabyte GA-8ITXE (based on the Intel 850GB) are in the 2.5 GBps range for both ALU/RAM and FPU/RAM benchmarks in SiSoft Sandra. This is about 20 per cent higher than the closest DDR competitor. The Gigabyte GA-8ITXE comes out on top scoring 2519 MBps in ALU/RAM and 2520 MBps in FPU/RAM tests. This proves beyond any doubt that RDRAM has no peer when it comes to pure memory bandwidth.

The DDR SDRAM boards based on the Intel 845D chipset came next, scoring in the 2 GBps range. It was a tough call in the DDR segment—the ASUS P4B266, MSI 845 Ultra-C and the MSI 845 Ultra-ARU were all neck and neck but the MSI 845 Ultra-C was just a shade faster with scores of 2044 MBps in the ALU/RAM and 1964 MBps in the FPU/RAM tests. The SiS 645 based ASUS P4S333 trailed the pack and only managed a paltry 1.8 GBps in the SiSoft Sandra memory benchmarks.

The conventional SDRAM boards trailed far behind as expected. Only Mercury's KOB845NFSX hit the 1000 MBps mark in the ALU/RAM benchmark. No other SDRAM-based board for the Pentium 4 even came close to this number!

Athlon XP/Duron: Coupled with the formidable Athlon XP 2000+ even the venerable VIA KT133A chipset using ordinary PC133 SDRAM showed some of its bite here. The Gigabyte GA-ZXR was the only KT133A board we received and it recorded some impressive scores in the SiSoft Sandra memory benchmark. In fact, both the Gigabyte entries (the other being the GA-7ZMMH based on the KM133A chipset) touched the magical 1 GBps throughput mark.

The DDR SDRAM based boards were in a different league altogether. The VIA KT333 based boards did not get a chance to shine as we used them with slower DDR266 memory modules (it will be some time before DDR333 memory debuts in India). Nevertheless, this is the first time we've seen a motherboard with support for a new memory standard without the memory itself being available in the market. However, the KT333 chipset is backward compatible and you can use DDR266 modules on these motherboards.

Surprisingly, the ASUS A7V266-E (using the older VIA KT266A chipset), came out on top with scores of 2037 MBps and 1888 MBps in the ALU and FPU tests. This board is so finely tuned that it even beats the KT333-based boards in pure memory throughput!

PentiumIII/Celeron (Tualatin): The variation in performance here is astonishing, with even the ASUS TUSI-M performing miserably, scoring only about 460 MBps which is about half of what its cousin, the TUSL2-M, achieved using the same memory and processor. The latest Intel 815EP used by this board is the best PC133 solution by far and the ASUS TUSL2-M squeezes every single drop of performance from it.

Disk subsystem

Performance within this subsystem depends on the motherboard's implementation of IDE channels. An underperforming IDE subsystem can cripple performance when it comes to video editing and other data-hungry applications. Each time these applications read or write data from or to the disk, instructions are passed to the drive controllers' device driver. The driver then pumps data through the PCI bus to the IDE controller. Improper implementation can severely hamper the entire data transfer process. If the PCI bus has not been iso-

Overclocker Friendly, Anyone?

Pushing the performance of hardware components has always excited PC enthusiasts, and we've never had it this good! Today, almost all motherboard manufacturers have some overclocking features built in, but some boards are simply an overclocker's dream come true.

The ASUS A7V333 is one such example. Supporting the latest DDR333 memory standard, you can boost your processor speed by adjusting the clock multipliers or the FSB speed. It even allows you to vary the CPU to RAM ratios and tweak the Vcore voltages. Imagine being able to push up the FSB speed from 133 MHz to a whopping 227 MHz in 1 MHz increments!

Almost all ASUS and MSI boards for the AMD processors have a decent amount of space around the CPU socket that let you install a larger than standard heatsink. The ASUS A7V266-E and the MSI KT3Ultra-ARU are prime examples of such overclocker friendly motherboard layouts. Also, ASUS has made it a point to ship motherboards with a minimum of three fan connectors, which comes in handy when you need extra cooling.

Soltek's SL75DRV4 is not too far behind. In addition to the usual FSB and CPU Vcore adjustments, it even lets you tweak the AGP and RAM voltage! Plus the BIOS implements the unique RedStorm overclocking feature, which automatically checks for the highest performance settings that can be made without compromising on stability and applies them without needing a system reboot.

While AMD boards have always been overclocker friendly, the Pentium 4 boards are not too far behind. MSI's 845 Ultra-ARU and the ASUS P4B266 allow FSB speeds up to 200 MHz, again in 1 MHz increments along with voltage adjustments for both Vcore and RAM.

The MSI 845 Ultra-ARU provides a hardware diagnostic device called the 'D-Bracket', which has four bi-colour LEDs. At reboot, these LEDs depict the POST (Power On Self Test) status and if any problems arise, they help to identify what has gone wrong. This is easier than the 'beep' error codes used by most motherboards today.